A Stability Calculus for Local Learning Rules

Abstract

This draft studies delete-one, add-one, replace-one, and leave-one-out perturbations for deterministic k-nearest neighbors on finite metric spaces with explicit tie-breaking. The final contribution language for the paper will be synchronized with the claim registry, proof notes, and novelty assessment so that theorem statements, computational certificates, and conjectural patterns remain clearly separated.

1 Related Work

Classical algorithmic stability studies how a learning rule changes when the training sample is perturbed, primarily in order to control generalization under an i.i.d. sampling model. The canonical reference is Bousquet and Elisseeff, who formalize several stability notions around replace-one perturbations and fresh-test evaluation points [1]. Kutin and Niyogi expand this landscape with weaker and almost-everywhere variants [11], while Feldman and Vondrák and later Bousquet, Klochkov, and Zhivotovskiy sharpen high-probability and rate-sensitive uniform stability bounds [8, 2]. Our paper does not compete with this line on sharper generalization bounds. Instead, we use it as the baseline vocabulary against which a more microscopic perturbation calculus can be compared.

1.1 Algorithmic stability and generalization

Most modern stability results are phrased around replacing one training example and then evaluating expected or fresh-test loss [1, 11, 14]. This perspective is powerful for generalization theory, but it can obscure a simpler definition-level point relevant for local rules: "change one sample point" is not a unique operation. Delete-one, add-one, replace-one, and leave-one-out share a family resemblance, yet they differ both in how the sample changes and in where the learner is evaluated afterward. Our contribution in this section is therefore not a new stability hierarchy in the sense of [1], but a deterministic finite-sample reframing that makes the perturbation operation and the evaluation point explicit.

1.2 Nearest-neighbor classification and deleted estimates

The nearest-neighbor line begins with Fix and Hodges and the classical risk results of Cover and Hart, together with the consistency framework later formalized by Stone [9, 5, 4, 18]. Those works study statistical performance under distributional sampling rather than worst-case single-perturbation effects. Closer to our setting are deleted-estimate and local-rule analyses such as Rogers and Wagner's finite-sample bound for local discrimination rules and the distribution-free deleted-estimate inequalities of Devroye and Wagner [17, 6, 7]. These are important precedents because they show that leave-one-out style evaluation has a substantial classical history around nearest-neighbor methods. Even so, their goal is not to separate LOO from adversarial replace-one at the level of fixed finite metric witnesses with explicit deterministic tie-breaking.

1.3 Leave-one-out, replace-one, and conformal stability certificates

Kearns and Ron connect algorithmic stability with leave-one-out cross-validation, already emphasizing that a CV-style perturbation has its own logic [10]. Recent conformal-prediction work makes related distinctions operational again, often in service of coverage certificates rather than 0–1 prediction loss. Examples include transductive stability analyses and training-conditional or stable conformal guarantees [15, 13, 16, 12]. We cite this line not because our paper is a conformal-prediction paper, but because it reinforces the message that leave-one-out style control and replacement-based control can lead to different guarantees. Our setting is more concrete and more local: deterministic k-NN on finite metric spaces with explicit top-k neighbor ordering.

1.4 Finite metric witnesses and local learning rules

Metric-space analyses of k-NN typically appear in consistency or geometric generality results rather than perturbation-separation examples [3]. We did not identify, in the sources inspected for this draft, a prior finite graph-metric example whose explicit purpose is to show that fixed-sample LOO stability need not control fixed-sample adversarial replace-one stability for deterministic 1-NN under the conventions used here. This absence is not enough to justify a sweeping novelty claim. It does, however, support a careful contribution statement: the paper's distinctive value lies in combining definition hygiene, explicit deterministic k-NN perturbation analysis, and a reproducible finite witness/certificate pipeline.

What this paper does not claim. We do not claim a new generalization bound, a contradiction to classical consistency results, or a theorem-level odd-k lifting result from current computational candidates alone. The paper's main ambition is narrower and, we think, cleaner: to make precise which one-sample perturbations are being discussed, how they interact at the uniform level, and how sharply they can separate at the pointwise finite-metric level for deterministic nearest neighbors.

2 Definitions and Preliminaries

2.1 Data Model and Sample

Definition 2.1 (Sample)**.** A *sample* S of size n is an ordered tuple

$$S = \big((x_1, y_1), \ldots, (x_n, y_n)\big)$$

where each $x_i \in X$ lies in a finite metric space (X, d) and each label $y_i \in \{0, 1\}$.

The order of the tuple is semantically relevant only for deterministic tie-breaking. Two identical labeled points appearing at different indices are treated as distinct sample occurrences.

- **Duplicate points are allowed.**
- **Conflicting labels at the same point are allowed.**

Any theorem or experiment that excludes duplicates or conflicts must state this explicitly.

Definition 2.2 (Finite Metric Space)**.** A finite metric space is a finite set X together with a distance function

$$d : X \times X \to \mathbb{R}_{\geq 0}$$

satisfying for all $x, x', z \in X$:

1. $d(x, x) = 0$;
2. $d(x, x') > 0$ for $x \neq x'$;
3. $d(x, x') = d(x', x)$;
4. $d(x, z) \leq d(x, x') + d(x', z)$.

Definition 2.3 (Graph Shortest-Path Metric)**.** For a finite, simple, connected, unweighted, undirected graph $G = (V, E)$, the graph metric is

$$d_G(u, v) = \text{length of a shortest path from } u \text{ to } v.$$

Disconnected graphs are rejected in code paths that require a metric.

2.2 Prediction Rule

The query point x may be any element of X, including a point that appears in the training sample.

Note 2.4 (Deterministic Tie-Breaking). Neighbor ordering is determined lexicographically by:

1. smaller distance to the query point;
2. smaller sample index in the ordered tuple.

If class votes tie after the first k ordered neighbors are selected, the predicted label is resolved by fixed label order $0 \prec 1$. Ties are always broken in favor of label 0.

Definition 2.5 (k-NN Prediction)**.** Given $k \in \mathbb{N}$ with $1 \leq k \leq n$, the deterministic k-NN predictor $h_S^{(k)}(x)$ is obtained by:

1. ordering the sample occurrences by the deterministic neighbor rule above;
2. selecting the first k occurrences;
3. taking the majority vote of their labels;
4. resolving label-vote ties by $0 \prec 1$.

2.3 Loss Function

Definition 2.6 (Binary 0-1 Loss)**.** For a hypothesis h and a labeled point $(x, y) \in X \times \{0, 1\}$, the binary classification loss is

$$\ell(h, (x, y)) = \mathbf{1}\{h(x) \neq y\}.$$

2.4 Sample Perturbations

Let $S = \big((x_1, y_1), \ldots, (x_n, y_n)\big)$.

Definition 2.7 (Delete-One)**.** For index $i \in \{1, \ldots, n\}$, define

$$S^{-i} = \big((x_1, y_1), \ldots, (x_{i-1}, y_{i-1}), (x_{i+1}, y_{i+1}), \ldots, (x_n, y_n)\big).$$

Definition 2.8 (Add-One)**.** For any labeled point $z = (x, y) \in X \times \{0, 1\}$, define

$$S \oplus z$$

as the ordered tuple obtained by appending z at the end.

Definition 2.9 (Replace-One)**.** For index i and labeled point $z = (x', y') \in X \times \{0, 1\}$, define

$$S^{i \leftarrow z}$$

as the ordered tuple obtained by replacing the i-th occurrence of S with z.

In v0.1 worst-case stability definitions, the replacement point ranges over all labeled points in $X \times \{0, 1\}$, not over a data-generating distribution.

2.5 Stability Notions

Unless otherwise stated, all pointwise notions are evaluated with fixed k, fixed metric space, fixed tie-breaking rule, and fixed loss defined above.

Note 2.10. Pointwise stability statements concern a fixed training sample, a fixed perturbation, and a fixed evaluation pair (x, y). Expected stability refers to the expectation under an explicitly specified data-generating distribution. Uniform stability is the supremum of the corresponding indicator over all valid ordered training tuples, all allowed perturbation choices, and all evaluation pairs.

Definition 2.11 (Pointwise Delete-One Stability Indicator)**.** For index i and query-label pair (x, y), define

$$\Delta_{\text{del}}(S, i, x, y) = \left|\ell(h_S^{(k)}, (x, y)) - \ell(h_{S^{-i}}^{(k')}, (x, y))\right|,$$

where $k' = \min(k, n-1)$ when deletion reduces sample size below k.

Definition 2.12 (Pointwise Replace-One Stability Indicator)**.** For index i, replacement z, and query-label pair (x, y), define

$$\Delta_{\text{rep}}(S, i, z, x, y) = \left|\ell(h_S^{(k)}, (x, y)) - \ell(h_{S^{i\leftarrow z}}^{(k)}, (x, y))\right|.$$

Definition 2.13 (Pointwise Add-One Stability Indicator)**.** For added labeled point z and query-label pair (x, y), define

$$\Delta_{\text{add}}(S, z, x, y) = \left|\ell(h_S^{(k)}, (x, y)) - \ell(h_{S\oplus z}^{(k)}, (x, y))\right|.$$

Definition 2.14 (LOO / CVloo Stability)**.** For each index i, define

$$\Delta_{\text{loo}}(S, i) = \left|\ell(h_{S^{-i}}^{(k')}, (x_i, y_i)) - \ell(h_S^{(k)}, (x_i, y_i))\right|.$$

LOO evaluates at the deleted sample occurrence itself. This definition is occurrence-based: if the same point appears twice, deleting one copy and evaluating at that occurrence is distinct from deleting the other.

For LOO, the uniform stability supremum is over ordered tuples S and indices i, with the evaluation pair fixed to the deleted occurrence.

2.6 Consistency Note

The consistency notion used in paper-level discussion is classical distributional consistency for k-NN classification. It is not a finite-sequence surrogate notion. Any finite-metric experiment provides evidence about worst-case stability behavior; it does not by itself constitute a consistency statement.

A Proof Sketches and Deferred Proofs

Proof to be inserted after Codex audit.

A.1 Uniform perturbation calculus

Proof to be inserted after Codex audit.

A.2 Deterministic k-NN perturbation criterion

Proof to be inserted after Codex audit.

A.3 Two-point 1-NN witness

Proof to be inserted after Codex audit.

B Enumeration Protocol

This appendix records the finite search spaces used for witness and certificate generation.

B.1 1-NN witness search

Proof to be inserted after Codex audit.

B.2 Tie-free filtering

Proof to be inserted after Codex audit.

B.3 Odd-k gadget candidate search

Proof to be inserted after Codex audit.

C Additional Figures

Additional publication figures are generated by repository scripts under ‘outputs/figures/‘.

Proof to be inserted after Codex audit.

D Bibliographic Notes

This appendix records scope notes for the literature review and citation boundary decisions.

Proof to be inserted after Codex audit.

References

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